

need for an infrastructure. The term 'piconet' is a collection of Bluetooth devices that are synchronised to the same hopping sequence. One device in the piconet can act as master and all other devices connected to the master act as slaves. The master determines the hopping pattern and the slaves have to synchronise to this pattern. The hopping pattern is determined by the device ID—a 48-bit worldwide unique identifier. The phase in the hopping pattern is determined by the master's clock. All active devices are assigned a 3-bit active member address.

All parked devices use an 8-bit parked member address. Devices in standby mode do not need an address. The goal for Bluetooth development was to use a single-chip, low-cost, radio-based wireless network technology for laptops, notebooks, headsets, etc.

Bluetooth operates in the 2.4GHz ISM band. However, MAC, physical layer and the offered services are completely different. Bluetooth transceivers use Gaussian FSK for modulation and are available in three power classes: Class 1 (max. power 100 mW), class 2 (max. power 2.5 mW) and class 3 (max. power 1 mW).

Journey from 1G to 4G

1G system. 1G specifications were released in 1990 to be used in GSM. 1G systems are analogue systems such as AMPS that use FDM to divide the bandwidth into specific frequencies that are assigned to individual calls.

2G system. These second-generation mobile systems are digital and use either TDMA or CDMA method. Digital cellular systems use digital modulation and have several advantages over analogue systems, including better utilisation of bandwidth, more privacy, and incorporation of error detection and correction.

2.5G system. It was introduced mainly to add latest bandwidth technology to the existing 2G generation. It supports higher-data-rate transmission for Web browsing and also supports a new browsing format language called wireless application protocol (WAP).

The different upgrade paths include high-speed circuit-switched data (HSCSD), GPRS and EDGE.

HSCSD increases the available application data rate to 14.4 kbps as compared to 9.6 kbps of GSM. By using four consecutive time slots, HSCSD is able to provide a raw transmission rate of up to 57.6 kbps to individual users.

GPRS supports multi-user network sharing of individual radio channels and time slots. Thus GPRS supports many more users than HSCSD but in a bursty manner. When all the eight time slots of a GSM radio channel are dedicated to GPRS, an individual can achieve as much as 171.2 kbps. But this has not brought any new evolution.

EDGE introduces a new digital modulation format called 8-PSK (octal phase-shift keying). It allows nine different air interface formats, known as multiple modulation and coding schemes, with varying degree of error control and protection. These formats are automatically and rapidly selectable. Of course, the covering range is smaller in EDGE than in HSCSD or GPRS.

3G system. To overcome the shortcomings of 2G and 2.5G, 3G has been developed. It uses a wideband wireless network that offers increased clarity in conversations. Countries throughout the world are currently determining new radio spectrum bands to accommodate 3G networks. ITU has established 2500-2690MHz, 1700-1855MHz and 806-960MHz bands. Here the target data rate is 2 Mbps. The data is sent through packet switching. Voice calls are interpreted through circuit switching.

3G W-CDMA (UMTS). Universal Mobile Telecommunication System (UMTS) or W-CDMA assures backward compatibility with 2G and 2.5G TDMA technologies. W-CDMA, which is an air interface standard, has been designed for always-on packet-based wireless service, so that computers and entertainment devices may all share the same wireless network and connect to the Internet anytime, anywhere.

W-CDMA supports data rates of up to 2.048 Mbps if the user is stationary,

thereby allowing high-quality data, multimedia, streaming audio, streaming video and broadcast type services to consumers. With W-CDMA, data rates from as low as 8 kbps to as high as 2 Mbps can be carried simultaneously on a single W-CDMA 5MHz radio channel, with each channel supporting between 100 and 350 simultaneous voice calls at once, depending on antenna sectoring, propagation conditions, user velocity and antenna polarisation.

Time slots in W-CDMA are not used for user separation but to support periodic functions. (This is in contrast to GSM where time slots are used to separate users). The bandwidth per W-CDMA channel is 4.4 to 5 MHz.

Since the global standard was difficult to evolve, three operating modes have been specified: A 3G device will be a personal, mobile, multimedia communication device (e.g., TV provider redirects a TV channel directly to the subscriber's phone where it can be watched). Second, it will support video conferencing, i.e., subscribers can see as well as talk to each other. Third, it will also support location-based services, where a service provider sends localised weather or traffic conditions to the phone or the phone allows the subscriber to find nearby businesses or friends.

3.5G. It supports a higher throughput and speed at packet data rates of 14.4 Mbps, supporting higher data needs of consumers.

4G system. It offers additional features such as IP telephony, ultra-broadband Internet access, gaming services and HDTV streamed multimedia. Flash-OFDM, the 802.16e mobile version of WiMax (also known as WiBro in South Korea), can support cellular peak data rates of approx. 100 Mbps for high-mobility communications such as mobile access and up to 1 Gbps for low-mobility communications such as nomadic/local wireless access, using scalable bandwidths of up to 40 MHz. The infrastructure for 4G is only packet-based (all-IP). ●

The author is director (R&D) at Global Institute of Technology, EPIP Sitapura, Jaipur